HGA2-FSL: A Heterogeneous Graph Aware-Aggregation Driven Few-Shot Learning Network for Hyperspectral Image Change Detection

Siyao Li, Xiaohan Xie, Xiaoyang Zhao, and Xianghai Wang

***Abstract*—Hyperspectral image (HSI) provides reliable data support for repeated observations of the overlap area through fine spectral reflectance data from numerous bands, enabling accurate capture of regional variation characteristics. However, the vast spectral and spatial information in HSI makes manual annotation laborious and complex. Practical applications thus demand models that achieve high performance with minimal labeled training data. Pay attention to this practical need, we propose a heterogeneous graph aware-aggregation driven few-shot learning (FSL) network (HGA2-FSL). This network enhances model representation through flexible graph construction and aims to improve the accuracy of hyperspectral change detection (CD) under limited labeling by transferring change priors learned from other images. Specifically, we first design a correlation aggregation module (CAM) to build multilevel graph structures that incorporate various types of geographic feature relationships, promoting effective capture and aware-aggregation of local and long-range information. In addition, through joint FSL on very-high-resolution image (VHRI) and HSI, the method utilizes the abundant labeled samples from VHRI and transfers learnable information, reducing the need for labeled samples of HSI. Meanwhile, we design a cross-domain node update-based domain adaptation (DA) method to explore a subspace that minimizes distribution differences, thereby reducing the modality gap between VHRI and HSI and improving the transfer learning performance. Extensive experiments on three public HSI-CD datasets validate the superior performance of HGA2-FSL. The code will be made publicly available at: https://github.com/lsylnnu/HGA2-FSL**

***Index Terms*—Change detection (CD), cross-domain, few-shot learning (FSL), graph neural network, hyperspectral image (HSI).**

I. Introduction

CHANGE detection (CD) is a geospatial analysis technique that rapidly identifies changes by examining the surface images captured over multiple periods [1]. This technology has become a research hotspot in various fields, including pollution monitoring [2], urban evolution [3], and natural ecological construction [4]. The continuous advancement of remote sensing (RS) imaging technology enables us to acquire a diverse range of RS images, such as very-high-resolution images (VHRIs), multispectral images (MSIs), and hyperspectral images (HSIs). Leveraging these rich data resources, researchers are developing an increasing number of CD methods.

Among the various types of RS images, HSI stands out due to its remarkable spectral resolution. Unlike other images, HSI not only provides precise geographic spatial information but also records data from hundreds of densely packed spectral bands within the visible to near-infrared range for each pixel [5]. Given that different materials have unique spectral reflectance characteristics, HSI can leverage this rich spectral information to provide strong support for more precise surface CD. Therefore, HSI-CD has emerged as a prominent topic in modern RS research. Recently, a significant amount of research has centered around the architecture of convolutional neural networks (CNNs). For instance, Ou et al. [6] constructed the relationships between each pixel and its spatial neighborhood pixels in the image using a CNN, thereby enhancing the feature representation of the pixels to be classified, which aids in pixel-level CD tasks. Feng et al. [7] developed a CNN architecture featuring adjustable kernels to precisely extract critical spectral features from HSIs while minimizing information redundancy. By enabling the extraction and integration of spectral features at suitable scales, this method significantly improves the ability to identify changes. Moreover, generative adversarial networks (GANs) are also a widely employed backbone architecture in HSI-CD. For example, Zhang et al. [8] proposed a generative cross-domain CD framework featuring a dual-branch generator and a three-branch discriminator. The generator synthesizes domain-shifted samples for data augmentation, while the discriminator extracts domain-invariant features, thereby bridging domain gaps and improving CD accuracy. It is noteworthy that CNN and GAN architectures are constrained by local receptive fields in processing HSI, thereby failing to model complex dependencies between long-range regions. Such global

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relationships are essential for a comprehensive understanding of intricate scenes in image data [9]. Although [10] introduced a global mixing module that brings distant pixels closer in the transformed layout by rearranging feature maps, thereby enlarging the receptive field of CNNs, such a random pixel reorganization may disrupt spatial continuity and introduce inconsistencies, making it difficult to model long-range relationships in a stable and fine-grained manner, motivating the need for more effective global relationship modeling.

To achieve the overall perception of the images, researchers have turned their attention to the transformer model. Originally developed for natural language processing, the transformer architecture is revolutionary in its ability to capture relationships between all elements in an input sequence, effectively identifying long-range dependencies [11]. Leveraging this, recent studies have introduced transformers into the HSI-CD field, achieving feature relationship capture over long distances and thus enhancing model performance. Wang et al. [12] used the transformer architecture to handle global spatial relationships and long-term temporal interactions in bi-temporal images, thereby enhancing the large-scale perception of image information. Chen et al. [13] employed the transformer to extend local perspectives to a global scale, learning both self-correlation and intercorrelation features, thereby more effectively distinguishing changed features from unchanged ones. Recently, to address the computational complexity limitations of the transformer in long-sequence processing, researchers have proposed the Mamba architecture, which features linear computational complexity and has since gained widespread adoption in HSI-CD [14]. Among them, Zhan et al. [15] employed Mamba as the core architecture to mine large-range and multiscale spatial–spectral features from HSI data, which combined frequency-domain analysis with prior knowledge to effectively enhance CD accuracy. Chen et al. [16] implemented global modeling and feature enhancement for entire bi-temporal images based on Mamba, thereby promoting precise perception of changes. Although transformer- and Mamba-based methods have made progress, they face a fundamental tradeoff between strongly data-driven modeling and the scarcity of labeled annotations in practical HSI-CD applications. Specifically, transformer and Mamba rely on global self-attention or selective scanning, both of which require adaptive inference of intertoken interactions from training samples. In essence, this is a data-dominated form of relational modeling that typically requires sufficient data to reliably identify, in a statistical sense, which long-range dependencies are truly critical. However, due to the inherent spectral–spatial complexity of HSI data and the high expertise required for accurate ground-truth annotation, acquiring high-quality labeled samples remains highly challenging. Consequently, when training data are limited, learning in transformer- and Mamba-based architectures can become unstable and prone to biased estimates, preventing their performance from being fully realized.

Therefore, there is an urgent need to explore an efficient architecture capable of modeling long-range dependencies while being more robust to the number of training samples. Graph convolution networks (GCNs) enable efficient learning through graph structures, maintaining long-range modeling capabilities while substantially reducing reliance on labeled samples, offering new possibilities for HSI-CD tasks [17]. Qu et al. [18] considered the flexible node aggregation capability of GCN and applied it to explore global contextual dependencies, thereby enhancing the ability of the model to extract global features and improve CD performance. Zhang et al. [19] exploited the strength of GCN in handling irregularly structured data and proposed a dual-branch GCN framework for CD, enabling large-scale feature extraction and aggregation to better capture subtle changes. However, existing GCN methods often rely on homogeneous or manually predefined simple graph structures, which may limit their adaptability to complex or heterogeneous relationships in images.

Therefore, the challenge of meeting the demands for effective long-range geospatial modeling while addressing the few-shot issue in HSI training remains formidable. In this article, we propose HGA2-FSL, a few-shot learning (FSL) network for HSI-CD, designed to perceive key information from different graph structure relationships and aggregate them, while significantly reducing the need for extensive training data. To comprehensively extract and leverage long-range semantic information from images, we design the correlation aggregation module (CAM). CAM constructs diverse graph structures based on different classes of image patches, coupling with an attention-guided aggregation processing mechanism to help the model learn long-range potential associations of features on a global scale. In addition, to optimize the learning process of the model in the case of scarce HSI annotation, we introduce VHRI data with simpler annotation and sufficient labeled samples, which aim to farthest utilize the learnable information from VHRI. By leveraging the joint FSL between VHRI and HSI, the ability of the model to learn change-related information from limited HSI annotations is maximized. This approach significantly reduces the adverse effects of the scarcity of HSI-labeled samples on model training.

Given the modality gap between VHRI and HSI (the cross-domain problem) that leads to negative outcomes in transfer learning, we propose a domain adaptation (DA) module. This module generates domain-wise graph structures based on a cross-domain node update strategy (cdNUP) to explore domain-invariant distribution spaces. Using it as a reference subspace for distribution alignment enables the model to explicitly model interdomain distribution discrepancy. Ultimately, DA is achieved by minimizing the distribution distance. By leveraging aligned multimodal features, we ensure that the learning process of the model is unaffected by modality differences. The main contributions of this work can be summarized as follows.

1) We propose a HGA2-FSL to effectively capture long-range feature relationships and enhance the ability of the model to perceive changes. By integrating FSL, this framework leverages well-labeled VHRI data alongside sparsely labeled HSI data, maximizing the utilization of learnable information to reduce the reliance on HSI annotations for model training.

2) A CAM is designed to create diverse graph structures by nodes from different classes. Guidance based on an attention mechanism, this module enables strongly correlated nodes to be aware of and aggregate each other, thereby effectively exploring and capturing both local and long-range relationships within the image.
3) The cdNUP is a novel DA method that constructs two cross-domain graph structures by employing the interaction updates between domain nodes, which are then used to generate a domain-invariant distribution space. Using this invariant space as a reference highlights interdomain differences, facilitating efficient alignment and overcoming the challenges posed by modality differences in cross-domain learning.

The remaining sections are summarized as follows. The relevant knowledge of GCN, FSL, and DA will be presented in Section II to facilitate a better understanding of HGA2-FSL. Section III provides a detailed description of the work addressed in this article. Section IV contains the experimental section, aimed at highlighting the performance of the model through detailed experimental results and analysis; and the critical elements of the experimental process, including specific parameter settings and their impacts, and provides an analysis of the DA visualization. Section V will summarize our work.

II. Related Work

A. GCN in HSI Processing

Unlike traditional CNNs, which struggle to explicitly model nonlocal spatial associations between nodes due to their fixed local receptive fields [20], and transformer architectures that incur high computational resource consumption when handling long-sequence data due to their global attention mechanisms [21], GCNs have natural adaptability to graph-structured data. This allows GCNs to automatically and efficiently construct spatial topological relationships within image scenes at low computational cost, effectively capturing the complex interactions and long-range dependencies among geographic features [22]. Leveraging these characteristics of GCNs, researchers have conducted extensive explorations in the field of HSI processing. For example, Dong et al. [23] combined CNN with GCN, leveraging the strengths of both architectures in feature capturing to enhance the classification performance of HSI classification networks. Qi et al. [24] utilized the graph relationships obtained from superpixel segmentation to learn better clustering representations, mitigating the negative impact of the contrastive learning process on the clustering structure. Zhang et al. [25] employed GCN to globally explore the band relationships in HSI, enabling the efficient selection of highly representative bands from a large number of bands to accomplish the band selection task successfully. Zhao et al. [26] aimed to organically combine transformer and GCN, focusing on change information within a global scale, so as to construct a more suitable network for CD tasks in HSI.

Based on graph structure construction and graph node aggregation, the aforementioned methods successfully capture long-range relationships between geographic features in images. This plays a positive role in comprehensively understanding scene information and facilitates downstream tasks. However, they still suffer from two limitations as follows.

1) The construction of graph structures relies on the results of superpixel segmentation, where the quality of segmentation directly determines the topological rationality of the graph structure and the representation ability of node features. Using a fixed superpixel segmentation method may introduce bias due to its inability to adapt to complex scenes, thereby limiting model performance.

2) For CD tasks, intraclass features exhibit more diverse expressions, while interclass features may show pseudo-similarity. Nonadjustable graph structures struggle to model the complex relationships within and between classes, leading to degraded performance when capturing subtle changes.

Inspired by GraphSAGE [27], we treat pixel-level features as graph nodes and analyze feature correlations to aggregate and update nodes, thereby achieving information perception and fusion from local details to global structures. In addition, by constructing variable multilevel graph structures, we establish diverse associations between different samples, providing effective support for deeply mining change information.

B. Cross-Domain FSL

FSL is an effective technique for addressing various tasks in data-scarce scenarios [28]. Its core objective is to enable efficient learning from a small number of training samples, allowing accurate reasoning and effective prediction of new feature instances in novel scenarios. Inspired by the ability of humans to leverage existing knowledge and experience to solve new problems quickly, FSL based on meta-learning employs meta-training strategies to enable models to extract useful information from limited data and rapidly adapt to new tasks based on the learned information [29].

Nowadays, in the field of HSI processing, many studies integrate FSL strategies with cross-domain tasks, aiming to leverage learnable knowledge from other RS images through cross-domain FSL. This approach further reduces the dependency on labeled hyperspectral data during model training [30]. As Li et al. [31] proposed, they achieved the transfer of learnable knowledge from the source domain to the target domain using self-distillation techniques and optimized this process with a designed distance metric. In addition, they employed a loss function to enhance the effectiveness of FSL strategies. This cross-domain FSL framework for HSI classification not only reduces the need for extensive labeled data in the target domain but also significantly improves the generalization capability of the model on new tasks. Ding et al. [32] developed an efficient local–global graph convolutional DA strategy and integrated it with cross-domain FSL, effectively supporting the generalization of source domain information to the target domain and achieving HSI classification under few-shot training conditions. Xi et al. [33] introduced a feature interaction module into a cross-domain FSL network, enhancing the expressive power and interaction of features from different domains during the FSL phase, thereby significantly improving the learning ability and classification performance of the model under few-shot conditions.

Hou et al. [34] developed an FSL framework for cross-scene HSI-CD, combining contrastive learning with FSL to alleviate label scarcity. It enhances discriminability via intradomain supervised contrastive learning and reduces domain gaps through cross-domain contrastive alignment, thereby improving detection performance under limited labeled samples. Feng et al. [35] proposed a fractional domain-based cross-domain FSL method for HSI-CD. It employs spatial–frequency joint learning to extract comprehensive discriminative representations of ground objects, models nonlocal topological relationships within each modality using GCN, and aligns cross-modal distributions through loss constraints to enhance the performance of cross-domain FSL.

In the field of RS image processing, CD requires experts to perform pixel-by-pixel analysis and labeling on multiple images. While the complexity of change patterns, combined with the need for domain-specific knowledge, makes label acquisition costly and time-consuming. This scarcity of training data heightens the demand for FSL methods. Although existing methods have achieved notable progress, CD still involves more complex feature representations and poses greater learning challenges. So, two key research challenges remain. First, how to effectively capture long-range geographic dependencies in RS scenes to extract robust and discriminative change features under few-shot conditions. And second, enhancing cross-domain information interaction to achieve better alignment and alleviate domain shift, thereby enabling the effective transfer of change priors in cross-domain FSL. To address these, we design CAM and cdNUP.

C. Domain Adaptation

DA is a common subtask in transfer learning, aimed at addressing the differences between cross-domain data (i.e., the feature distributions of the source domain and target domain) during knowledge sharing to improve model performance on the target domain [36]. Many DA methods have emerged in the field of HSI processing. For example, Huang et al. [37] designed a feature interaction generator to continuously fit and generate domain-invariant features while applying a discriminator to further minimize the distribution discrepancy of the generated features. Huang et al. [38] utilized a hybrid convolution-based generator to extract features and constructed a three-branch discriminator to constrain cross-domain feature distributions toward consistency from multiple dimensions. Qu et al. [39] introduced a pseudo-label cyclic strategy, which iteratively selects and updates pseudo-training samples in the target domain to progressively reduce the feature representation gap between the source and target domains. Xu et al. [40] developed a transformation mechanism that maps target data to a projection space based on specified rule criteria within the mechanism, thereby assisting the source domain in approximating a feature distribution more similar to that of the target domain. Li et al. [41] implemented a distance metric loss function at the final stage of feature extraction to optimize the training process and ultimately reduce the differences in feature distributions across domains.

In this article, we propose a novel DA method, cdNUP, which explores a subspace with minimal distribution discrepancy through the interactive aggregation of graph node features between the source and target domains. This subspace provides a credible reference for the model to learn the difference between domains, and finally realizes the DA by minimizing the difference, which reduces the feature distribution gap between different domains and enhances the cross-domain learning efficiency under limited labeled samples.

III. Methodology

This section provides a detailed introduction to the HGA2-FSL framework. First, an overview of network architecture is presented. Then, we describe each component in detail, including the CAM, the implementation of cross-domain FSL, and the cdNUP. Finally, we explain the defined loss function used to optimize the model.

A. Overall Framework Design of HGA2-FSL

The proposed HGA2-FSL framework architecture is depicted in Fig. 1, which takes inputs from both the source domain and target domain. Among them, VHRI with sufficient labels is used as the source domain data, and HSIs with limited labels are used as the target domain data. First, the bi-temporal images undergo differencing and patching processing to obtain sets of difference image patches: $X_V = \{x_i^V \mid i = 1, \ldots, U\} \in \mathbb{R}^{p\times p\times BV}$ and $X_H = \{x_j^H \mid j = 1, \ldots, M\} \in \mathbb{R}^{p\times p\times BH}$, where, x_i^V is the ith VHRI differential patch, x_j^H is the jth HSI differential patch, U and M represent the number of the two types of patches, respectively, $p \times p$ is the spatial size of the patch, BV is the number of VHRI bands, and BH is the number of HSI bands. Subsequently, five labeled samples from each class in X_H are randomly selected and augmented by adding Gaussian noise. Finally, the network training is conducted using these augmented samples along with 500 randomly selected samples per class from X_V.

During the training phase following aspects include.

1) First, the input data passes through the unified module shown in Fig. 1 and is convolution processed via two layers of 2D-CNN to extract the band-dimensional features, which are reduced to 100 dimensions, so as to be uniformly processed by the subsequent shared CAM module. The CAM module then facilitates both local and long-range information propagation. Among them, the local features extracted by CNN-Group are transformed into graph node features, and the multilevel heterogeneous graph structure is constructed under the guidance of the attention mechanism. Then, the node features within the heterogeneous graph are aggregated using GCN (refer to Section III-B for details).
2) The features output from the previous stage are divided into support sets and query sets by class for cross-domain FSL, aiming to fully utilize the learnable change information from the source domain data and fully exploit the medium-to-long-range information from HSIs (refer to Section III-C for details).
3) Finally, the DA module based on cross-domain node update explores a subspace in which domain-invariant

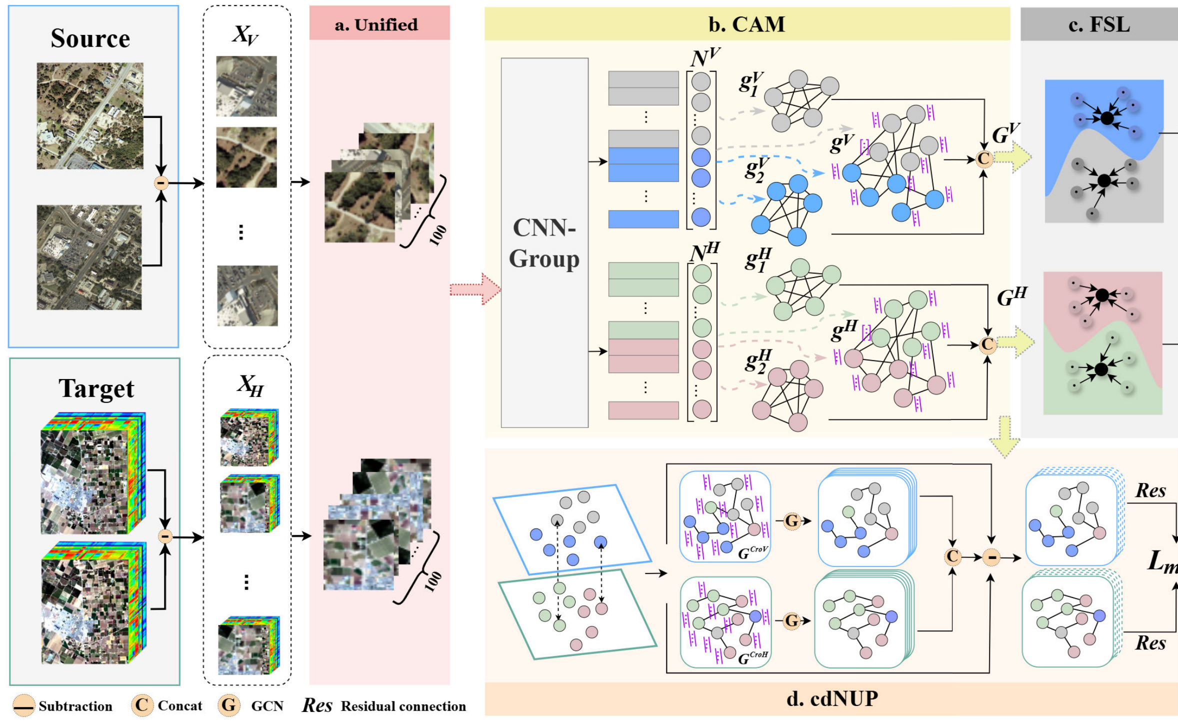


Fig. 1. Architecture of proposed HGA2-FSL, which consists of four core components: (a) unified module, which aligns the band dimensions of source and target domain imagery to establish a unified feature space for subsequent shared module processing. (b) CAM serves as a feature extraction component that constructs multilevel heterogeneous graphs to integrate local neighborhood features and global long-range dependencies. (c) FSL module, which executes cross-domain FSL. (d) cdNUP, which facilitates efficient alignment and overcoming the challenges posed by modality differences in cross-domain learning.

feature distribution information provides powerful reference information for capturing interdomain differences. Finally, DA is achieved by minimizing the difference in feature distribution (refer to Section III-D for details).

B. Correlation Aggregation Module

To address the limited receptive field of CNN and enable the model to utilize the features captured at a larger scale to better understand and represent the change information, a novel CAM is designed. It integrates CNN with GCN (using GraphSAGE as the GCN layer), where the organic combination of the two realizes the flexible capture and utilization of both local and nonlocal information in the image.

First, extract the local information through CNN-Group, which consists of multiple sets of 3D-CNNs organized in a residual structure. This module processes the training data from both the source domain and target domain after dimension unification, performing patch-level local feature extraction. This process can be formulated as

$$\begin{cases} n_i^V = RE\left(x_i^{\prime V}\right) \\ n_j^H = RE\left(x_j^{\prime H}\right) \end{cases} \tag{1}$$

where $x_i^{\prime V} \in \mathbb{R}^{9\times 9\times 100}$, $x_j^{\prime H} \in \mathbb{R}^{9\times 9\times 100}$, $RE(\cdot)$ are CNN-Group, and n_i^V and n_j^H are the features finally extracted by CNN-Group. The detailed network structure specifications are shown in Table I, where bz is the batch size, N indicates that the corresponding operation is skipped, and Y indicates that the operation is performed.

TABLE I
CNN-GROUP OVERALL STRUCTURE

No.	Layer name	Filter size	Output size	Padding	Normalization and activation
1	Input	-	(bz,9,9,100)	-	-
2	Expand dims	1	(bz,1,9,9,100)	N	N
3	3D conv layer 1	(3,3,3,8)	(bz,8,9,9,100)	Y	Y
4	3D conv layer 2	(3,3,3,8)	(bz,8,9,9,100)	Y	Y
5	3D conv layer 3	(3,3,3,8)	(bz,8,9,9,100)	Y	Y
6	No. 3 + No. 5	-	(bz,8,9,9,100)	-	-
7	Max pooling 1	(2,2,4)	(bz,8,5,5,25)	N	N
8	3D conv layer 4	(3,3,3,16)	(bz,16,5,5,25)	Y	Y
9	3D conv layer 5	(3,3,3,16)	(bz,16,5,5,25)	Y	Y
10	3D conv layer 6	(3,3,3,16)	(bz,16,5,5,25)	Y	Y
11	No. 8 + No. 10	-	(bz,16,5,5,25)	-	-
12	Max pooling 2	(2,2,4)	(bz,16,3,3,7)	N	N
13	3D conv layer 7	(3,3,3,32)	(bz,32,1,1,5)	N	N
14	Reshape layer	-	(bz,160)	-	-

The CNN-Group-extracted local features undergo vectorization through a transform dimension layer (flatten operation), yielding 1-D feature vectors that serve as graph nodes. Thus, these local features are used to construct class-wise and batch-wise graph structures. For 1 batch input, the node set can be expressed as

$$\begin{cases} N^V = \{n_1^V, n_2^V, \ldots, n_{r/2}^V, n_{r/2+1}^V, \ldots, n_r^V\} \\ N^H = \{n_1^H, n_2^H, \ldots, n_{l/2}^H, n_{l/2+1}^H, \ldots, n_l^H\} \end{cases} \tag{2}$$

where N^V and N^H denote the node sets of the source domain and target domain, respectively, $r/2$ is the number of nodes of each class in N^V, and $l/2$ is the number of nodes of each class in N^H. Note that the CD task can be regarded as a binary

	n_1^V	n_2^V	n_3^V	n_4^V	...	$n_{r/2}^V$
n_1^V	0	1	1	1	1	1
n_2^V	0	0	1	1	1	1
n_3^V	0	0	0	1	1	1
n_4^V	0	0	0	0	1	1
...	...	...	...	...	...	...
$n_{r/2}^V$	0	0	0	0	0	1

Fig. 2. Class-wise adjacency matrices of source domain subgraphs.

classification task, where the two classes are the variable class and the invariant class.

Unlike the random sampling in GraphSAGE to select nodes, this study posits that nodes of the same class exhibit strong correlations, while nodes of different classes still maintain certain underlying relationships. Therefore, the attention mechanism is used to guide the selection of relevant nodes.

First, construct the class-wise graph structure. Samples from the same class are highly correlated. The connections and aggregation between them help capture more discriminative features, enabling the network to focus more on key information. Therefore, for 1 batch of input data, nodes of the same class are selected as adjacent nodes, and edges are added between them. According to this connection method, a class-wise subgraph containing the nodes of the same class and the edges between them is constructed for each class. The edge information (adjacency matrix) of this graph is shown in Fig. 2 (taking the subgraph constructed with the first class of samples in the source domain as an example), where 1 indicates an edge connection between two nodes, and 0 indicates no connection. Finally, $g_c^V (c = 1, 2)$ denotes the class-wise subgraph of the source domain, with c representing the cth class, and g_c^H denotes the class-wise subgraph of the target domain. Subsequently, a batch-wise graph structure is constructed, that is, a graph is built for the nodes with 1 batch of input. The attention score matrix between nodes is calculated through the attention mechanism. According to this matrix, the previous k nodes with the highest attention score for each node are regarded as its most relevant neighbors. Therefore, a batch-wise graph structure with rich long-distance semantic information is constructed through the guidance of the attention mechanism. Specifically, for 1 batch of input N^V, the query and the key for the calculation of the attention score matrix are obtained through linear transformation, which are denoted as Q^V and K^V. Subsequently, the calculation process of the attention score matrix can be expressed as

$$AS^V = Softmax\left(\frac{Q^V \left(K^V\right)^T}{\sqrt{d}}\right) \tag{3}$$

where Softmax is the normalization function, $(K^V)^T$ represents the transpose of K^V, d represents the feature dimension of the node, and AS^V is the attention score matrix obtained from the source domain data.

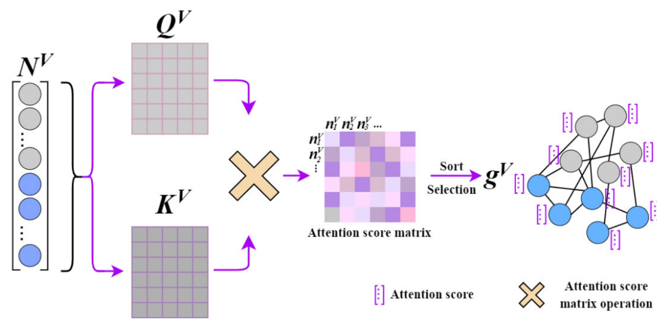


Fig. 3. Batch-wise graph construction process.

Similarly, the attention score matrix AS^H of the target domain can be obtained

$$AS^H = Softmax\left(\frac{Q^H \left(K^H\right)^T}{\sqrt{d}}\right). \tag{4}$$

In the matrix AS^V and AS^H, the value at the position (a, b) represents the correlation score between the ath node and the bth node. Based on this, for each row of the attention score matrix, the column indices corresponding to the K highest values are selected; that is, the most relevant nodes for each node are identified, and edges are established between them. Finally, the constructed batch-wise graph for the source domain is denoted as g^V and the batch-wise graph of the target domain is denoted as g^H. Fig. 3 takes the construction of the source domain batch-wise graph as an example to show the attention-guided construction process.

Finally, perform nonlocal information aggregation. Based on the constructed multilevel heterogeneous graphs g_c^V, g_c^H, g^V, and g^H, respectively, through two layers of GraphSAGE convolutional layers with a mean aggregation function, the features of the corresponding nodes are calculated using their neighboring information of the nodes. Ultimately, the graph features obtained can be expressed as G_{f1}^V, G_{f2}^V, G_{f1}^H, G_{f2}^H, G_F^V, and G_F^H, respectively. Among them, G_{f1}^V is the feature derived from g_1^V, G_F^V is the graph features derived from g^V, and the rest is the same. Subsequently, the features from different levels of the graphs are integrated. For the source domain, the process can be expressed as

$$G^V = Drop\left(Conv2d\left(CAT\left(CAT\left(G_{f1}^V, G_{f2}^V\right), G_F^V\right)\right)\right) \tag{5}$$

where $CAT(\cdot)$ is the connected operation, $Conv2d(\cdot)$ is a 2D-CNN layer, and $Drop(\cdot)$ is the drop operation. The final output features G^V of the source domain are obtained by the integration of multilevel heterogeneous graph features through this hierarchical fusion pipeline.

Similarly, the final output features G^H for the target domain can be expressed as

$$G^H = Drop\left(Conv2d\left(CAT\left(CAT\left(G_{f1}^H, G_{f2}^H\right), G_F^H\right)\right)\right). \tag{6}$$

In CAM, the combination of CNN and GCN enables flexible capture of local to nonlocal information in an image. Among them, CNN effectively extracts the local features from Euclidean data through convolution operations. Then, GCN further processes and aggregates these features in the non-Euclidean space, utilizing the long-range dependencies to learn complex representations on a larger scale. Furthermore, in the constructed multilevel heterogeneous graph structure, the implementation of class-wise subgraphs and attention-guided batch-wise graphs enables each node to participate in diverse relational contexts. This architecture facilitates richer information representation and extraction of correlated features exhibiting both intraclass discriminative equilibrium and interclass semantic relationships. It not only enhances the capacity of the model to comprehend inherent complex patterns in HSIs, but also improves the CD performance in FSL scenarios of the model.

C. Cross-Domain FSL Scheme

The graph features G^V and G^H obtained through multilevel heterogeneous graph feature integration are used for cross-domain FSL. The implementation proceeds as follows: first, based on G^V and G^H, the node features are selected by class and divided into a support set and query set for FSL. Specifically, for each class, randomly select nodes to form the support set, with the remaining nodes constituting the query set. That is, $S^V = \{S_1^V, \ldots, S_c^V\}$, $Q^V = \{Q_1^V, \ldots, Q_c^V\}$, among them, S^V is the set of source domain support sets, S_c^V represents a subset of the support set for class c, Q^V is the set of source domain query sets, and Q_c^V denotes a subset of the query set for class c. Similarly, the target domain support and query sets can be obtained $S^H = \{S_1^H, \ldots, S_c^H\}$, $Q^H = \{Q_1^H, \ldots, Q_c^H\}$. Subsequently, cross-domain FSL is achieved by calculating the Euclidean distance between the features of nodes in each query set and the prototype representation of the corresponding support set. Specifically, during each iteration, FSL is simultaneously performed for both source and target domains. Taking the source domain as an example, the formula can be expressed as

$$P\left(y = c \mid q_{c_r}^V \in Q_c^V\right) = -eu\left(Mean\left(S_c^V\right), q_{c_r}^V\right) \tag{7}$$

where $Mean(\cdot)$ is the calculation formula for the prototype representation of the cth class support set, $eu(\cdot,\cdot)$ represents the Euclidean distance metric, $q_{c_r}^V$ is the rth node in Q_c^V, y is the class label, and $P(\cdot)$ is the predicted probability output for the query set samples.

Similarly, the predicted probability for the target domain query set can be obtained as

$$P\left(y = c \mid q_{c_r}^H \in Q_c^H\right) = -eu\left(Mean\left(S_c^H\right), q_{c_r}^H\right). \tag{8}$$

During the training process, FSL is conducted simultaneously on both the source and target domains, achieving full utilization of learnable change information from the source domain, comprehensive extraction of discriminative information from HSIs, and effective cross-domain learning with limited labeled samples.

D. DA Scheme Based on Cross-Domain Node Update

The source domain and target domain data used in HGA2-FSL are cross-domain data captured by different sensors in different surface cover scenarios, resulting in different

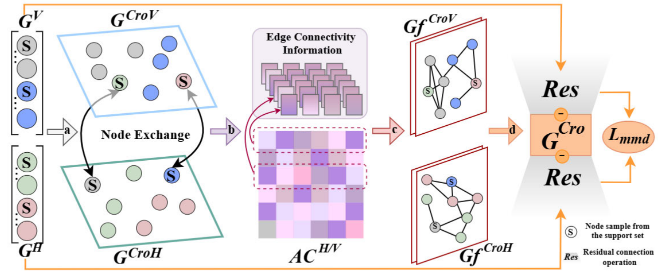


Fig. 4. cdNUP flowchart can be divided into four steps, corresponding to arrows a, b, c, and d. (a) By swapping the support set nodes, cross-domain node sets G^{CroV} and G^{CroH} are obtained. (b) Attention weight matrices $AC^{H/V}$ are computed to determine edge connectivity information for constructing cross-domain graph structures. (c) GCN is performed to propagate and update features, and then the features of two domains are fused to extract cross-domain shared distribution features G^{Cro}. (d) Using G^{Cro} as a reference subspace for distribution alignment, differential operations and residual connections are applied between G^{Cro} and the original domain features to explicitly model interdomain distribution discrepancies. Finally, DA is achieved by minimizing the distribution distance.

feature distributions. This distribution inconsistency in the data will affect the performance of cross-domain learning in the model. Based on this, we design a DA module that relies on cross-domain node update to effectively capture the common feature distribution across domains, thereby constructing a domain-invariant subspace that highlights critical interdomain distinctions. Ultimately, by systematically minimizing the distribution discrepancy between domains, effective DA is achieved. The overall framework of the proposed DA scheme is illustrated in Fig. 4.

1) Obtaining Cross-Domain Node Set: Based on the cross-domain FSL scheme described in Section III-C, support set S^V and query set Q^V can be divided from the G^V, while support set S^H and query set Q^H are correspondingly obtained from the G^H. Building upon this foundation, cross-domain node features can be obtained by implementing support set swapping between domains. Specifically, the support sets of source and target domains are exchanged and then combined with the query sets within their respective domains to form new node feature sets. This swapping mechanism effectively breaks the intradomain feature distribution independence and establishes relationships between cross-domain samples through interaction. The cross-domain node sets can be formulated as follows:

$$\begin{cases} G^{CroV} = \{S^H, Q^V\} \\ G^{CroH} = \{S^V, Q^H\} \end{cases} \tag{9}$$

where G^{CroV} represents the cross-domain node set of the source domain after node exchange and G^{CroH} represents those of the target domain.

2) Constructing Cross-Domain Graph: After obtaining cross-domain node sets for both domains, we further calculate intradomain node affinities using a process consistent with 3. Taking the source domain as an example, the cross-domain node features in the G^{CroV} are first linearly projected via learnable weight matrices to generate the query and key parameter matrix. The affinity computation follows the scaled dot-product mechanism, which involves an inner product operation between the query matrix and the key matrix, where the inner product is normalized by the square root of the key dimension to mitigate gradient saturation. Subsequently, the result of this process is passed through the $softmax$ function to obtain normalized attention weights AC^V, which quantify the relevance between each node pair and establish a probabilistic distribution for subsequent graph aggregation. The target domain follows the identical procedure, producing attention weights AC^H.

For the attention weight matrix, the ith row explicitly encodes the relevance of the ith node to all other nodes. By performing top-K selection via row-wise sorting, we identify the K most related nodes for each target node. The process provides edge connectivity information by identifying the most relevant nodes for each node, which enables the construction of a topological structure with adaptive connectivity, referred to as the cross-domain graph. Specifically, the cross-domain graph for the source domain is denoted as g^{CroV}, while the graph for the target domain is represented as g^{CroH}.

3) Extracting Cross-Domain Shared Distribution Feature: Ultimately, the aggregation and updating of node features are achieved through GCN, and cross-domain shared representations are obtained through dual domain spatial fusion

$$\begin{cases} Gf^{CroV} = GCN_2\left(g^{CroV}\right) \\ Gf^{CroH} = GCN_2\left(g^{CroH}\right) \\ G^{Cro} = CAT\left(Gf^{CroV}, Gf^{CroH}\right) \end{cases} \tag{10}$$

where $GCN_2(\cdot)$ denotes two graph convolutional layers. G^{Cro} is the cross-domain shared distribution feature.

Overall, the aforementioned process through cross-domain node interaction enables the model to learn the distributional characteristics of one domain while incorporating supportive information from the other, thereby effectively enhancing its capacity to model common features of interdomain. Specifically, through affinity-based search between cross-domain support sets and query sets, followed by node feature aggregation and update: on one hand, an attention-guided neighborhood selection mechanism establishes topological consistency constraints; on the other hand, dynamic feature propagation reinforces semantic alignment, ultimately forming a cross-domain associative representation. Finally, by fusing the updated graph features of the source and target domains in the feature space, the resulting interdomain graph features effectively capture the shared semantic topology between the two domains, reflecting their intrinsic common distributional characteristics. This provides a referential domain-invariant subspace for subsequent processes aimed at reducing interdomain distribution discrepancy.

4) Minimizing Feature Distribution Differences: Based on the above process, the cross-domain node update method captures the shared feature distribution between the source and target domains, yielding a subspace G^{Cro} with the minimum distribution difference. Subsequently, subtracting G^{Cro} from G^V and G^H, followed by residual connections, enhances the individuality of the feature distribution within each domain. This approach effectively highlights interdomain distribution differences

$$\begin{cases} GF^{CroV} = Res\left(\left(G^V - G^{Cro}\right), G^V\right) \\ GF^{CroH} = Res\left(\left(G^H - G^{Cro}\right), G^H\right) \end{cases} \tag{11}$$

where $Res(\cdot)$ denotes the residual connection operation, GF^{CroV} represents the individuality feature distribution of the source domain, and GF^{CroH} represents the individuality feature distribution of the target domain.

Ultimately, the maximum mean discrepancy (MMD) [42] was used to measure the difference in feature distribution between the source domain and the target domain data. Therefore, the MMD distance between GF^{CroV} and GF^{CroH} can be expressed as

$$\begin{aligned} &MMD^2\left[\mathcal{F}, GF^{CroV}, GF^{CroH}\right] \\ &= \left\| \frac{1}{n^V}\sum_{r=1}^{n^V} f\left(gf_r^{CroV}\right) - \frac{1}{m^H}\sum_{l=1}^{m^H} f\left(gf_l^{CroH}\right)\right\|_H^2 \end{aligned} \tag{12}$$

where $\mathcal{F}$ is the set of mapping functions in the regenerated kernel Hilbert space, gf_r^{CroV} is the rth data point in GF^{CroV}, and gf_l^{CroH} is the lth data point in GF^{CroH}.

E. Design of Loss Function

The source domain FSL and the target domain FSL are carried out simultaneously in each training iteration. Based on the prediction probability of the samples in the support set, the FSL loss function of the source domain, derived from the cross-entropy function, is defined as

$$L_{fsl}^V = -\sum_{(q,y)\in Q_c^V} logP\left(y = c \mid q\right). \tag{13}$$

Similarly, the target domain FSL function is given by

$$L_{fsl}^H = -\sum_{(q,y)\in Q_c^H} logP\left(y = c \mid q\right). \tag{14}$$

For the DA loss, we use the MMD distance [see (15)] to compute the discrepancy between the source and target domains

$$\begin{aligned} L_{MMD} = &\frac{1}{n^{V^2}}\sum_{r=1}^{n^V}\sum_{r'=1}^{n^V}\varphi\left(gf_r^{CroV}, gf_r^{CroV'}\right) \\ &- \frac{1}{n^V m^H}\sum_{r=1}^{n^V}\sum_{l=1}^{m^H}\varphi\left(gf_r^{CroV}, gf_l^{CroH}\right) \\ &+ \frac{1}{m^{H^2}}\sum_{l=1}^{m^H}\sum_{l'=1}^{m^H}\varphi\left(gf_l^{CroH}, gf_l^{CroH'}\right) \end{aligned} \tag{15}$$

where $\varphi(\cdot)$ is the kernel function.

Ultimately, the total loss function for HGA2-FSL combines the above terms

$$L_{ALL} = L_{fsl}^V + L_{fsl}^H + \alpha L_{MMD} \tag{16}$$

where α is a hyperparameter that balances the contribution of DA within the overall framework. The selection of its value will be discussed in Section IV-D.

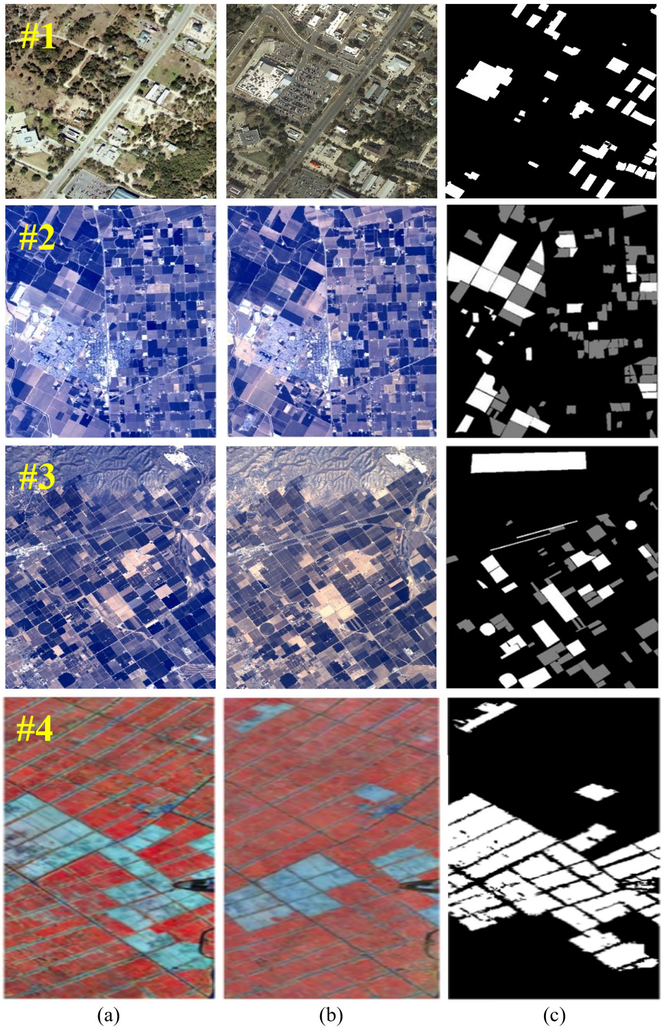


Fig. 5. Schematic of the source and target domain datasets. (a) Image acquired on T1. (b) Image acquired on T2. (c) Ground-truth image.

TABLE II
COMPARE EXPERIMENTAL RESULTS

Method	Bay Area		Santa Barbara		Farmland	
	OA	Kappa	OA	Kappa	OA	Kappa
ML-EDAN	0.7804	0.4346	0.8393	0.6488	0.8003	0.5311
CSANet	0.7825	0.5295	0.8502	0.6395	0.9141	0.7962
SSFTT	0.6868	0.3774	0.7905	0.5617	0.8595	0.6706
S3Net	0.6562	0.3179	0.8234	0.6241	0.8655	0.6560
DCFSL	0.8665	0.7287	0.8622	0.6973	0.9248	0.8139
Gia-CFSL	0.9008	0.7991	0.9199	0.8251	0.9308	0.8418
HGA2-FSL	**0.9470** ↑ 4.62%	**0.8931** ↑ 9.4%	**0.9430** ↑ 2.31%	**0.8793** ↑ 5.42%	**0.9513** ↑ 2.05%	**0.8807** ↑ 3.89%

IV. EXPERIMENT AND ANALYSIS

A. Dataset and Experimental Setup

The experiment adopted the LEVIR-CD dataset (#1) [43], a VHRI dataset for CD research. A pair of temporal images is randomly selected from this dataset as source domain data. In addition, three widely used public HSI-CD datasets are also introduced as target domain data to evaluate the performance of the proposed method in HSI-CD: the Bay Area dataset (#2) [44], the Santa Barbara dataset (#3) [44], and the Farmland dataset (#4) [45]. Fig. 5 shows the bi-temporal pseudo-color images and corresponding ground-truth images of the source and target domain datasets.

HGA2-FSL is implemented based on the Pytorch framework and the NVIDIA GTX 3080 Ti GPU. The optimizer selects adaptive moment estimation (Adam), the learning rate is set to 0.001, and the learning rounds are 2000. For training, 500 labeled samples per class are selected from the source domain, while 5 samples per class are drawn from the target domain. During cross-domain FSL, the support and query sets are partitioned as follows: 1 sample per class is allocated to the support set, and 15 samples per class to the query set.

HGA2-FSL is evaluated against six state-of-the-art methods. Multilevel encoder–decoder attention network (ML-EDAN) [46], intertemporal interactive attention network (CSANet) [47], space-spectrum feature vector transformer network (SSFTT) [48], spectral–spatial twin network (S3Net) [49], deep cross-domain FSL network (DCFSL) [50], and graph information aggregation cross-domain FSL network (Gia-CFSL) [51].

The performance of HGA2-FSL and comparison methods is quantified using overall accuracy and the Kappa coefficient, calculated as follows:

$$\mathrm{OA} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}} \tag{17}$$

$$\begin{cases} \mathrm{P_{TP}} = (\mathrm{TP}+\mathrm{FP}) \times (\mathrm{TP}+\mathrm{FN}) \\ \mathrm{P_{TN}} = (\mathrm{TN}+\mathrm{FP}) \times (\mathrm{TN}+\mathrm{FN}) \\ \mathrm{PE} = \dfrac{\mathrm{P_{TP}}+\mathrm{P_{TN}}}{(\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN})^2} \\ \mathrm{Kappa} = \dfrac{\mathrm{OA}-\mathrm{PE}}{1-\mathrm{PE}} \end{cases} \tag{18}$$

where TP, TN, FP, and FN, respectively, represent the correctly predicted changed pixels, the correctly predicted unchanged pixels, the misclassified unchanged pixels, and the misclassified changed pixels.

B. Comparative Experiments and Analyses

Table II summarizes the OA and Kappa coefficient achieved by HGA2-FSL and six comparison methods on three datasets. Fig. 6 comparatively illustrates the CD results obtained by different methods on these datasets. All comparative experiments are conducted under identical sampling rates to HGA2-FSL for fair comparison. Specifically, the noncross-domain method utilized five labeled samples per class for training the target domain hyperspectral data, while the cross-domain method employed VHRI data with 500 labeled samples per class to train the target domain classifier, thereby maintaining experimental consistency and comparability.

On the Bay Area dataset, HGA2-FSL achieved the highest OA and Kappa values. Compared with the Gia-CFSL method, which obtained the best results among the six comparison methods, HGA2-FSL achieved significant improvements of 4.62% in OA and 9.4% in Kappa. Combined with the CD result map and ground-truth map in Fig. 6 for analysis, the blue rectangle region in Fig. 6 is complex and changeable,

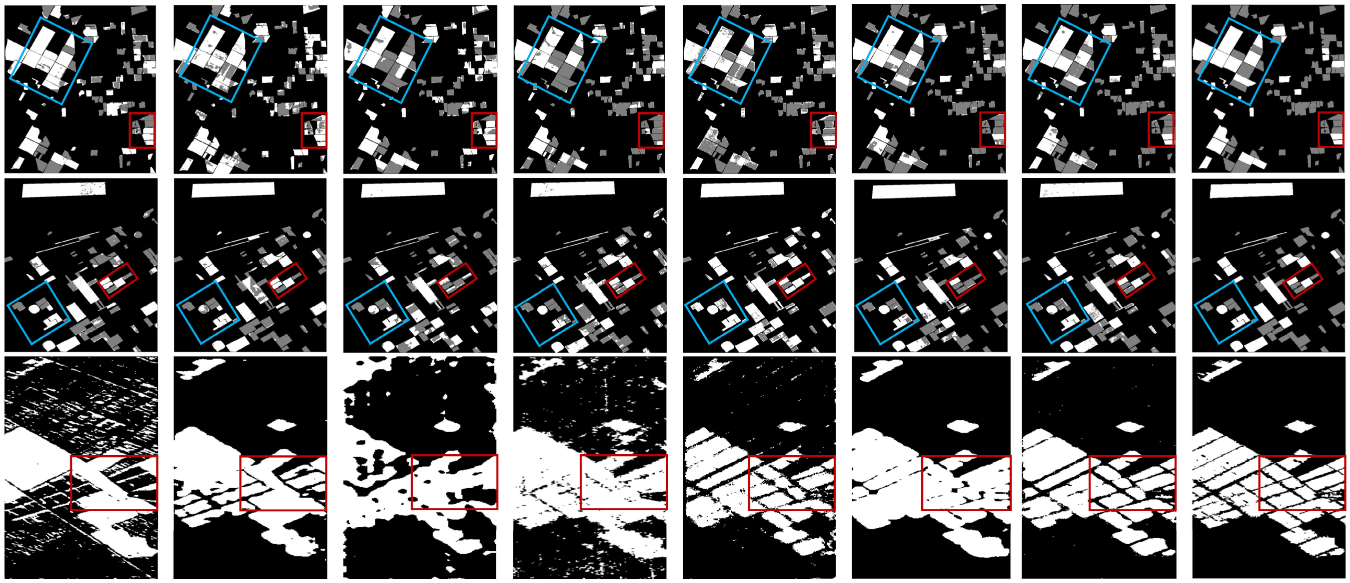


Fig. 6. Prediction results on three datasets. (a) ML-EDAN. (b) CSANet. (c) SSFTT. (d) S3Net. (e) DCFSL. (f) Gia-CFSL. (g) HGA2-FSL. (h) Ground-truth.

and the changed and unchanged parts within the area are distributed alternately, which increases the difficulty of CD. In this region, methods such as ML-EDAN, CSANet, SSFTT, and S3Net have made relatively serious errors in distinguishing between the changed and unchanged categories. While the two FSL-based methods, DCFSL and Gia-CFSL, achieved relatively accurate CD overall, their accuracy still fell short of that of our proposed HGA2-FSL approach. The CD results of HGA2-FSL in the blue rectangular area are closer to the ground-truth map. Within the red rectangular region, HGA2-FSL simultaneously achieves excellent CD performance. The comparative results clearly demonstrate HGA2-FSL's dual capability to precisely identify change areas while effectively suppressing false alarms, even in challenging scenarios with complex spatial patterns.

On the Santa Barbara dataset, the inherent class imbalance between changed and unchanged categories presents significant challenges for CD tasks. AS evidenced in Fig. 6, the red rectangle belongs to a difficult-to-detect area. The ML-EDAN, CSANet, SSFTT, DCFSL, and Gia-CFSL algorithms all have extensive false detectives and missed detectives. S3Net, as a method dedicated to solving the few-shot problem, has achieved relatively accurate results in this area. However, in contrast, HGA2-FSL can achieve superior performance in this area, which is nearly consistent with the ground-truth map. This confirms that the method proposed in this article has better CD performance. Through visual comparison of the detection results in the blue rectangular area, shows that HGA2-FSL achieves optimal CD results. Regarding the two quantitative indicators, HGA2-FSL achieved the highest indicator value for both OA and Kappa. Moreover, compared with the suboptimal results, it demonstrated 2.31% improvement in OA and 5.42% improvement in Kappa.

On the Farmland dataset, the ground-truth map in Fig. 6 shows that the boundaries between the changed and unchanged

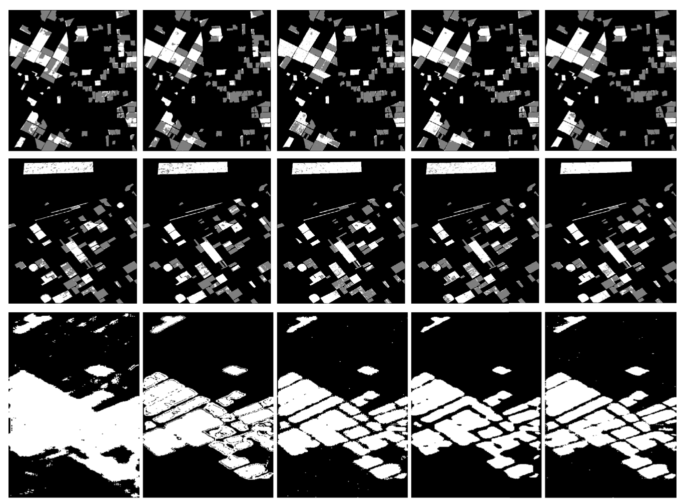


Fig. 7. Ablation results on the three datasets. (a) Baseline. (b) W-G^2. (c) WO-DA. (d) WO-FSL. (e) HGA2-FSL.

regions are interlaced, and the edges of each region are discontinuous. The analysis reveals that methods such as CSANet, S3Net, and Gia-CFSL have great difficulty in accurately identifying the mixed unchanged areas within the changed areas of this dataset. The edges of the changed areas predicted by methods such as ML-EDAN, CSANet, and SSFTT are smooth, and there are missed detections in the unchanged areas. Furthermore, methods such as ML-EDAN, SSFTT, S3Net, and DCFSL incorrectly classify some pseudo-change areas as changed areas within large unchanged areas. In contrast, the HGA2-FSL proposed in this article can better distinguish between changed and unchanged areas in the Farmland dataset, thereby reducing both false and missed detections. This improved performance is also evident in

TABLE III

ABLATION EXPERIMENTAL RESULTS

	a	b	c	d	e
Ablation model	Baseline	W-G^2	WO-DA	WO-FSL	HGA2-FSL
dataset	Bay Area				
OA(%)	86.14	88.38	92.26	90.59	**94.70**
Kappa(%)	71.93	76.43	84.39	81.14	**89.31**
dataset	Santa Barbara				
OA(%)	84.37	88.51	91.95	88.99	**94.30**
Kappa(%)	68.49	74.42	82.59	76.42	**87.93**
dataset	Farmland				
OA(%)	88.22	93.23	94.29	93.62	**95.13**
Kappa(%)	74.42	83.67	84.06	84.10	**88.07**

the area within the red rectangle in the result map, where HGA2-FSL has achieved a more refined CD. From the analysis of OA and Kappa values, the method in this article also achieved the best values. Compared with the best-performing comparison method, OA is further improved by 2.05%.

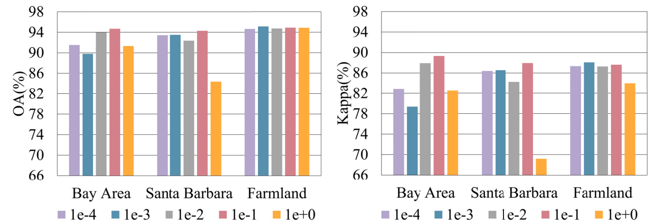
C. Ablation Experiment and Analysis

HGA2-FSL employs a multilevel spatiospectral feature extraction framework based on CAM. The method effectively utilizes learnable change information from the source domain and fully exploits discriminative information in HSIs under limited labeled sample conditions through cross-domain FSL. In addition, it addresses feature distribution discrepancies between domains and achieves feature distribution alignment via a cross-domain node update mechanism for DA. To validate the importance and effectiveness of each component in the proposed framework, the ablation experiments are conducted. The results of five ablation models are presented in Table III, which are: a) baseline: A CNN-Group-based feature extraction framework serving as the baseline; b) W-G^2: CD using CAM-based feature extraction; c) WO-DA: HGA2-FSL without the DA component; d) WO-FSL: HGA2-FSL without the cross-domain FSL module; and e) HGA2-FSL: the complete proposed framework. The corresponding visual ablation results are shown in Fig. 7.

Analysis of OA and Kappa values in Table III (columns a and b) demonstrates that the multilevel graph structure construction significantly enhances CD accuracy across all three datasets. It can be proved that the baseline CNN-Group architecture alone is insufficient for extracting deep data features and learning essential data attributes. The proposed CAM-based approach with diverse graph connections improves local detail preservation nonlocal long-distance dependency capture, and change information representation capability.

The ablation study in column c of Table III reveals that removing the cross-domain node update module in HGA2-FSL degrades performance, demonstrating that DA is crucial for VHRI-to-HSI scenarios, and feature distribution alignment facilitates effective cross-domain learning. Similarly, column d in Table III shows that excluding cross-domain FSL, substantially reduces OA and Kappa values, proving that FSL enables effective learning from limited labeled data, facilitates cross-domain knowledge transfer, and optimizes information utilization in VHRI and HSI applications.

The complete HGA2-FSL consistently outperforms all ablation variants in CD results, confirming that: 1) each component contributes synergistically to performance; 2) the integrated framework achieves optimal accuracy; and 3) all elements are essential for robust cross-domain CD.



(a) (b)

Fig. 8. Impact of hyperparameters on CD performance. (a) OA achieved by HGA2-FSL across different α values. (b) Kappa obtained by HGA2-FSL across different α values.

D. Hyperparameter Analysis

The HGA2-FSL framework incorporates an adjustable hyperparameter α in the DA loss component of its total loss function to regulate the DA effect. The evaluated values for α are $\{1e-4, 1e-3, 1e-2, 1e-1, 1e+0\}$. Fig. 8 presents the OA and Kappa values achieved across the three datasets (Bay Area, Santa Barbara, and Farmland) under different α values. The hyperparameter values have varying degrees of influence on the model's CD performance. Overall, the Bay Area and Santa Barbara datasets show strong sensitivity to α variations, while the Farmland dataset demonstrates relatively stable performance across α values. To achieve optimal performance, we set the α values to $1e-1$, $1e-1$, and $1e-3$ for the Bay Area, Santa Barbara, and Farmland datasets, respectively.

E. DA Performance Analysis

HGA2-FSL addresses the fundamental challenge of processing heterogeneous-domain data, where the source and target domains originate from different ground-covering scenarios and sensor systems. This heterogeneity creates significant discrepancies in feature distributions in the embedding space, known as domain shift. The existence of this problem will hinder the transfer of learnable information in the source domain and further affect the efficiency of cross-domain FSL. Our solution incorporates a novel DA module featuring cross-domain node updates, specifically designed to minimize feature distribution differences between domains. Fig. 9 presents the T-SNE visualizations comparing feature distributions before and after DA. This visualization demonstrates that our DA module effectively bridges distribution differences and promotes consistency between heterogeneous-domain data. In Fig. 9, the red dots represent source domain data while the green dots represent target domain data. Fig. 9(a) shows

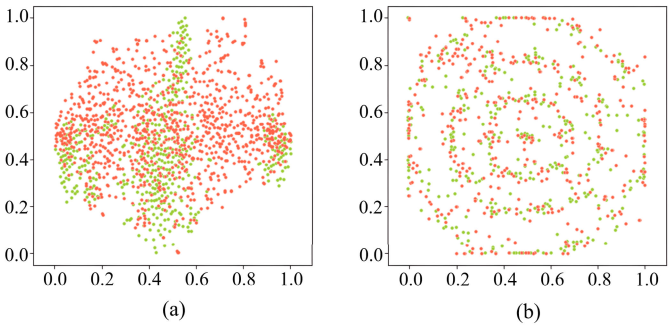


Fig. 9. Feature distribution comparison for the Farmland dataset. (a) Original feature space before DA. (b) Aligned feature space after DA.

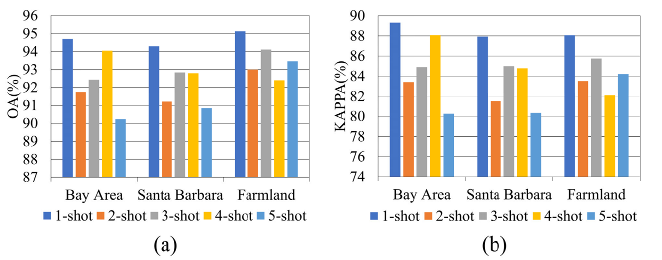


Fig. 10. Effect of different support set sample size. (a) Value of OA. (b) Value of Kappa.

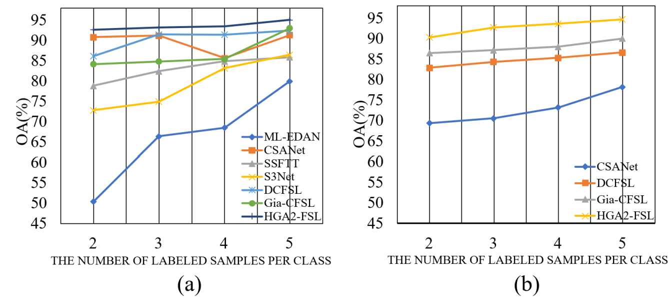


Fig. 11. CD results under different numbers of labeled samples. (a) On the Farmland dataset. (b) On the Bay Area dataset.

the preadaptation feature distribution, revealing significant distribution differences in the $[0, 1]$ space, where the source domain data are widely distributed in space, while the distribution of the target domain is concentrated in three strip regions. Fig. 9(b) shows the postadaptation results, in which we observe significant alignment between the source and target domain distributions and near-complete overlap between the two domains. These results provide strong evidence that our DA module successfully aligns feature distributions, reduces discrepancies between heterogeneous domains, and facilitates cross-domain FSL.

F. Effect of Different Support Set Sample Size

To investigate the impact of support set size on the proposed cdNUP strategy, we conducted experiments across three datasets with the number of support samples ranging from 1-shot to 5-shot. The results are shown in Fig. 10. Analysis indicates that more support samples do not necessarily lead to better performance. This phenomenon can be explained by the synergistic mechanism between cross-domain FSL and DA. In the cross-domain FSL stage, the model constructs class prototypes from the feature means of the support set samples, enabling query samples to move closer to their corresponding prototypes for classification. By exchanging support set nodes, cdNUP aims to break the relatively closed feature distributions within each domain. This allows the source and target domains to not only learn their own distributions but also receive guidance from the support set of each other's domain, thereby driving the cross-domain distributions closer together and progressively extracting more generalizable common features. However, an increase in support set size does not always lead to performance improvements. Excessive support samples may introduce intradomain redundancy or noise, blurring the focus of cross-domain alignment and even exacerbating interdomain distribution differences. Experiments show that under the 1-shot setting, the node exchange strategy performs optimally across all three datasets. This configuration avoids redundant information and efficiently promotes the learning of cross-domain common features, leading to better DA outcomes.

G. CD Results Under Different Numbers of Labeled Samples

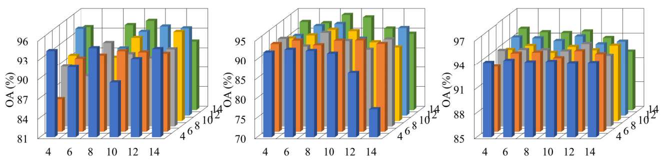
To evaluate the impact of the number of labeled samples in the target domain on the CD performance, we designed experiments with varying configurations of labeled samples. Given the graph-based architecture of HGA2-FSL, which requires at least two nodes to establish connections, we initiated experiments with the minimum setting of 2 labeled samples per class. Fig. 11 presents the OA results of different methods on the Farmland dataset under configurations of 2–5 labeled samples per class. Experimental analysis reveals a strong positive correlation between OA and labeled sample quantity, validating that supervised signals remain critical for model learning. Among all comparative methods, HGA2-FSL consistently achieved superior performance across diverse few-shot sampling rates, demonstrating its effectiveness for few-shot CD tasks. To further validate the generalization capability of HGA2-FSL, we selected several top-performing methods on Farmland and conducted comparative experiments on the Bay Area dataset. Results further confirm the robust performance of HGA2-FSL under few-shot conditions.

H. Analysis of the K-Value Selection

In both CAM and cdNUP, a top-K selection mechanism is used to filter the neighbor nodes that participate in graph aggregation for each node. We set a moderate range of K values, {4, 6, 8, 10, 12, 14}, based on a fixed batch size of 32. Specifically, the graph is constructed within each training batch, where each node selects its top-K neighbors from the other samples in the same batch according to the learned affinity scores. Thus, K is naturally bounded by the batch size, and we keep K below half of the batch size to preserve sparsity and avoid overly dense, redundant aggregation. Too small K may yield under-connected graphs with limited message passing, whereas too large K may introduce noisy or

TABLE IV
COMPUTATIONAL COMPLEXITY COMPARISON OF DIFFERENT METHODS

Method		ML-EDAN	CSANet	SSFTT	S3Net	DCFSL	Gia-CFSL	HGA2-FSL
Params(M)		223.13	2.39	0.15	0.15	4.26	0.31	4.68
Training time(s)	Bay Area	417.74	249.17	7.01	14.52	381.91	325.43	63.86
	Santa Barbara	542.44	449.65	13.62	27.21	515.26	464.22	62.74
	Farmland	243.02	151.73	6.88	11.74	293.71	473.89	63.19



(a) (b) (c)

Fig. 12. Analysis of the K-value selection. (a) On the Bay Area dataset. (b) On the Santa Barbara dataset. (c) On the Farmland dataset (x-axis is K1 and the y-axis is K2.)

irrelevant neighbors and increase computational overhead, so we adopt a moderate range for a practical tradeoff between efficiency and effectiveness. Fig. 12 presents the experimental results of different K-value combinations on the three datasets, where K1 and K2 correspond to the K settings in CAM and cdNUP, respectively. The optimal configurations determined are: [K1 = 8, K2 = 4] for the Bay Area dataset, [K1 = 8, K2 = 14] for the Santa Barbara dataset, and [K1 = 12, K2 = 8] for the Farmland dataset.

I. Number of Parameters and Training Time

As shown in Table IV, there are noticeable differences in the number of parameters and training cost among the compared methods. ML-EDAN has 223.13-M parameters, representing a typical heavily parameterized model. In contrast, SSFTT and S3Net have only 0.15-M parameters, making them extremely lightweight but also limiting their expressive capacity. Our method has 4.68-M parameters, placing it in the same order of magnitude as DCFSL. It remains significantly smaller than heavyweight networks while avoiding the representation deficiency that may arise from extremely small models. Notably, training time does not scale linearly with the parameter count, as it is also influenced by the computational pattern and the number of training iterations. Despite having more parameters than CSANet and Gia-CFSL, HGA2-FSL requires less training time on the three datasets, remaining around 63 s and consistently lower than most competing methods. This suggests that our approach provides a more favorable balance between model capacity and training efficiency.

V. CONCLUSION

In this article, an FSL HSI-CD method based on heterogeneous graph aware-aggregation is proposed. While addressing the problem of limited labeled samples in HSIs, the rich spatial–spectral features of HSIs are fully explored through the heterogeneous graph structure to mine key change information. Specifically, a CAM module is designed first to guide the construction of the multilevel graph structure by embedding the attention mechanism. This process not only includes the extraction of local features of a node, but also involves the perception of more complex long-distance dependencies. In multilevel heterogeneous graphs, through attention guidance, the same node can aggregate important semantic information across different structural scales, enhancing the model's ability to represent changing information. Furthermore, in order to make full use of the learnable information and reduce the reliance on labeled samples, a cross-domain FSL strategy is adopted. By implementing this strategy on VHRI data with sufficient labels (source domain) and hyperspectral data with limited labels (target domain), the model extracts common features from the source domain and adaptively transfers them to the target domain. In addition, an innovative DA method is proposed, which seeks a subspace minimizing distribution differences through cross-domain node updates, achieving precise capture of discrepancies and promoting knowledge transfer. Experiments on three widely used CD datasets verify the effectiveness of the proposed method. Under the challenging few-shot condition (five labeled samples per class), the proposed method achieves optimal OA and Kappa scores compared to recent advanced algorithms, demonstrating stability in detection performance.

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